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Smart Bike Theft Tracker

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1 Introduction & Requirements

The goal of this project is to design the hardware aspect of an anti bike theft device, that has the capability to detect motion, pinpoint bike location using GPS and transmitting an alert and bike coordinates via LoRa to the bike owner in case of theft. The device must also include a way to arm and disarm the device (e.g. a switch) to avoid false theft detection, and the device must also include an ultra low power mode to limit current draw while it is stationary or in the unarmed state.

We will implement motion detection using an accelerometer, that has the capability of producing an interrupt when movement is detected. This will be used to wake the MCU up from the deep sleep state in case the bike is armed and is stolen.

To arm and disarm the device, we will use a slide switch as a proof of concept. In practice a slide switch is too simple and obvious for a device of this nature, but implementing a different arming solution is not complicated (NFC would be a good alternative for a device like this). While the device is armed, the interrupt from the accelerometer will tell the MCU that the device is being stolen, and while the device is unarmed the MCU will stay in sleep mode.

A GPS will track the position of the device, and in case of a theft the MCU will get the current coordinates from here. The GPS will require an antenna to establish a connection to satellites. In case of a theft, the owner of the bike will receive an alert on their mobile device through LoRa along with the current coordinates of the bike.

2 System Architecture

A block diagram of the Smart Bike Theft tracker can be seen in figure 2.1. It shows each major block of the system with key components under each block. The peak current draw of key blocks is also included to provide an overview of total current draw and the balance of power draw of each major block.

The power supply block is powered through a USB type-C terminal to comply with EU legislation. A battery charger IC uses the 5 V from the USB to charge the battery and to drive the Switch Mode Power Supply, which powers the rest of the system with 3.3 V.

The accelerometer draws 140 μA of current at peak and will send an interrupt to the MCU block when movement is detected. The MCU can communicate with the accelerometer through I2C. The MCU draws about 7 mA at peak current draw. This includes the draw of the LoRa module as it is included in the MCU chip.

The GPS block will have the largest current draw at peak, but the 67 mA is a worst case scenario over several GPS modules in NEO-6 family. It will likely be lower in practice. In total the peak current draw will be about 74 mA.

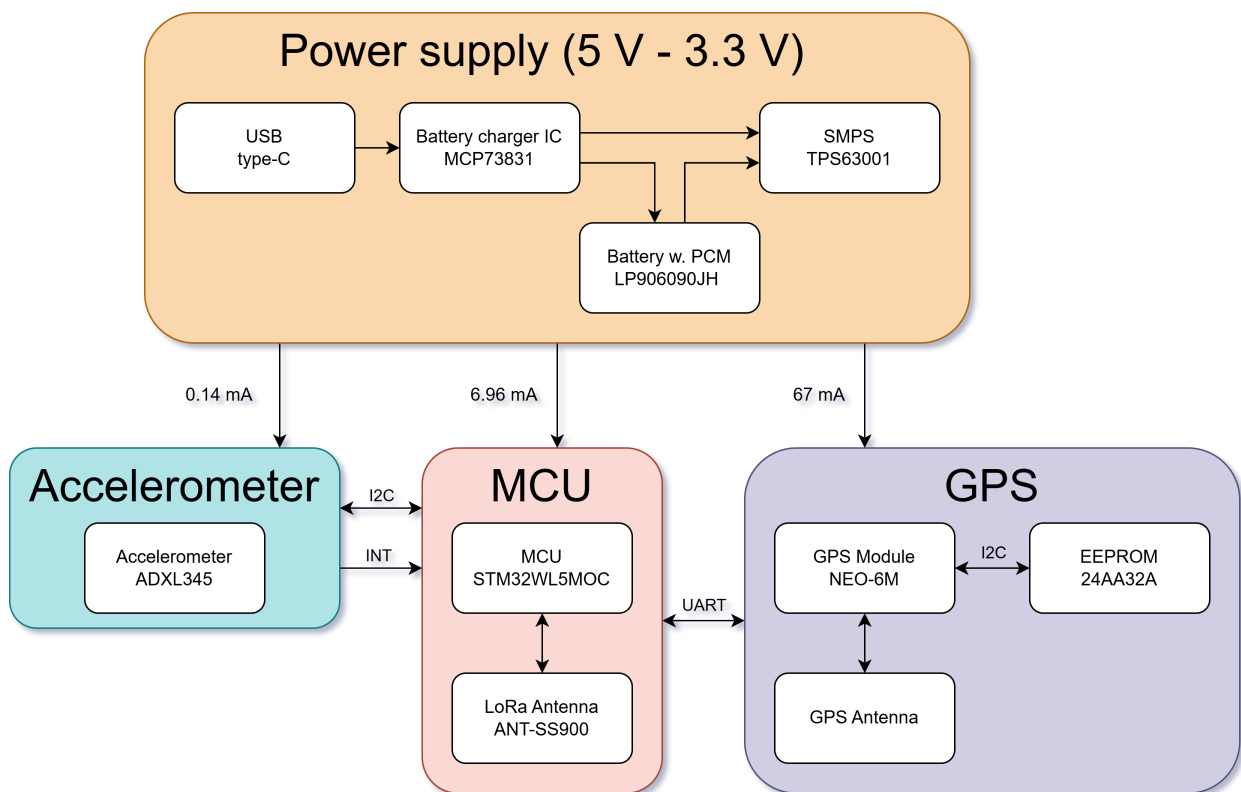


Figure 2.1: Block diagram of the Bike theft device

3 Component Choices

This chapter explains the choices that were made for each of the components used in the Smart Bike Theft Tracker design.

3.1 MCU & LoRaWAN

For the MCU we have chosen the STM32WL5MOCH6TR from STMicroelectronics, which has an integrated LoRaWAN module. This choice of MCU has a significant downside of costing 12.43 EUR, which is more expensive than other popular microcontrollers from STM that are in the price range of 5-10 EUR, but this downside is outweighed by many upsides.

The biggest upside comes from the integrated LoRaWAN module, which decreases design complexity and size of the PCB layout. The STM32WL5MOCH6TR is equipped with internal oscillator networks eliminating the need for external crystals. The matching network for the LoRaWAN antenna along with an RF filter is not necessary in the design as it is also included in the MCU package. Because of this, the smaller PCB will result in a smaller product all together and make the Smart Bike Theft Tracker easier to hide on a bicycle, making the product more effective.

The STM32WL5MOCH6TR covers all requirements from our design conditions. That being I2C, SPI and UART communication support, interrupt pins that can wake the MCU from sleep mode, and a sleep mode with very low current draw. The second upside comes with the manufacturer, as STM has very good documentation of their products, and where other cheaper manufacturers might omit information in datasheets, STM is more likely to have it in theirs. STM also has very good support as many people use their products, and because of the open nature of the documentation, users are able to help each other with troubleshooting.

The upsides to the STM32WL5MOCH6TR, more than makes up for the larger cost.

3.2 Accelerometer

The ADXL345BCCZ-RL7 is a good choice for the Smart Bike Theft Tracker, even though it is more expensive than many basic accelerometers.

Its main advantage is that it can be connected to the MCU's WKUP1 pin (PA0), so the sensor can wake the system directly from low-power sleep. Combined with its small 3 mm × 5 mm package, I2C/SPI support, FIFO buffer, and built-in motion features such as tap, activity, inactivity, and free-fall detection, it reduces PCB size, firmware complexity, and power consumption.

The higher unit cost is therefore justified by the design simplification it provides. Instead of continuously polling raw acceleration data, the tracker can rely on the ADXL345 to detect

events locally and only wake the MCU when needed, which makes the final product more compact and more battery efficient.

3.3 GPS

The NEO-6M-0-001 GPS module is a practical choice for the Smart Bike Theft Tracker, though its main drawback is that the NEO-6 family has reached End-of-Life, making it less ideal for long-term new designs. Despite this, it remains attractive because it is a self-contained GPS receiver with a simple UART interface that can be connected directly to the MCU without additional protocol conversion hardware. The module communicates over UART using TXD and RXD pins, with a default baud rate of 9600 bps that is sufficient for standard NMEA output and keeps the interface straightforward to implement and debug. Physically, the module measures approximately $16\text{ mm} \times 12.2\text{ mm} \times 2.4\text{ mm}$, which keeps the PCB compact and leaves room for the battery and radio section. It operates from 2.7 V to 3.6 V, and many breakout versions accept 3 V to 5 V input, simplifying integration with common MCU boards. For this project, the NEO-6M-0-001 reduces design complexity by providing ready-to-use position data over serial, eliminating the need for the host processor to handle navigation signal processing, which simplifies firmware and shortens development time. In a theft tracker, this is valuable because the GPS subsystem should remain small, reliable, and easy to power-manage by waking only when needed.

3.4 Battery charger IC

We have chosen the MCP73831-2ACI/OT battery charger IC from Microchip and the LP605060JU lithium polymer battery from Jauch Quartz as the charging and battery solution. The charger IC is designed for both lithium ion and lithium polymer battery chargers and the battery charge voltage is also compatible with the charger IC at 4.2 V. The battery has a built in protection circuit module, keeping it from discharging the voltage too low or charging it too high and for preventing high currents in the battery.

The battery was chosen for its size, as it closely matches the dimensions of the PCB. This will minimize the final size of the theft tracker, making it easier to hide. The battery has a capacity of 1900 mAh.

We have chosen the TPS63001 as the switch mode power supply as it will be effective in converting the battery voltage to the correct 3.3 V rail needed for the MCU and peripherals. Calculations for the power supply solution will be carried out in the next chapter.

4 Power Design

We will choose between an LDO and an SMPS to drive the 3.3 V power rail of our design. We will do this based on the efficiency, the quiescent current and the complexity tied to the solution. We will start by looking at an LDO for which we have gone with the MCP1700, which is a popular LDO that is capable of regulating voltages from 1.2-5 V. The input of the LDO must be at least $(3.3 \text{ V} + 3\%) + V_{DROPOUT}$. The dropout voltage of the MCP1700 is 350 mV max. This gives a minimum input of 3.749 V, which is already larger than the nominal voltage of a LiPo battery cell. Assuming the battery can hold the minimal voltage necessary for operation, and the max current is drawn from the LDO, the power loss can be calculated.

$$P_{loss} = (3.749 \text{ V} - 3.3 \text{ V}) \cdot 74 \text{ mA} = 33.226 \text{ mW} \quad (4.1)$$

The MCP1700 has a max quiescent current of 4 μA at no load. Despite the downsides it has very low complexity, as only two capacitors are needed apart from the LDO itself.

Looking at the chosen TPS63001 SMPS we see the minimum input is 1.8, well below the minimal battery voltage of 3 V. The efficiency of the SMPS can be seen in figure 4.1. We see at an output current of 100 mA, the efficiency is around 90% at the voltage range of the LiPo battery. This corresponds to a loss of 27.133 mW. The quiescent current when no load is applied to the SMPS is up to 51.5 μA . This is a lot more than the LDO. Finally, the complexity of the TPS63001 is more intricate, as more components and different types, like an inductor, are necessary.

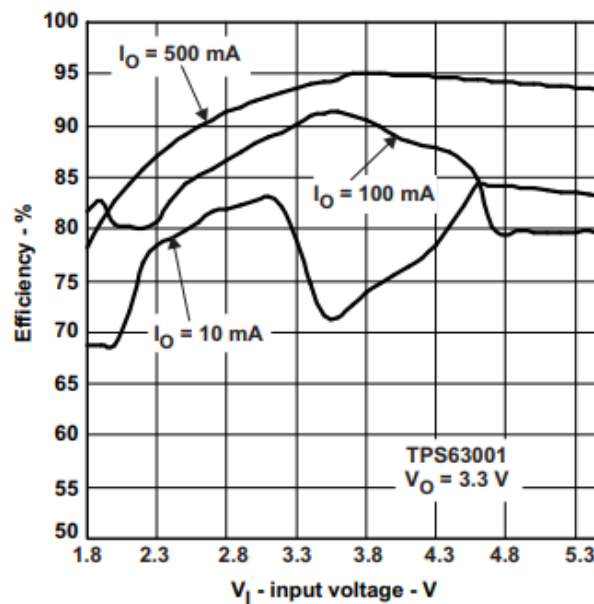


Figure 4.1: Efficiency of the TPS63001 SMPS versus the input voltage.

To conclude the comparison, the LDO beats the SMPS in quiescent current and complexity, but the SMPS beats the LDO with a small margin in efficiency, however the LDO will not operate optimally once the battery dips below the minimum LDO input voltage. Because of this we have chosen the TPS63001 SMPS as the driver of the 3.3 V rail.

As stated in an earlier chapter, the peak current draw of the MCU, accelerometer and GPS is about 74 mA, which is satisfied by the SMPS as it has a max current output of at least 1.2 A at the battery voltage of around 3.0-3.7 V input. See figure 4.2.

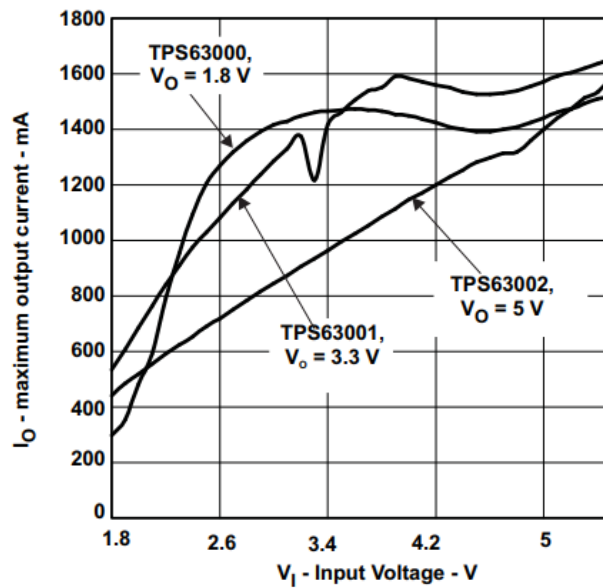


Figure 4.2: Output current of the TPS63001 SMPS versus the input voltage.

The output power from the SMPS is $3.3 \text{ V} \cdot 74 \text{ mA} = 244.2 \text{ mW}$. If we assume the efficiency of the SMPS to be 80% in the worst case scenario and the input voltage at the minimal battery voltage, the input current will be $244.2 \text{ mW} / (0.8 \cdot 3 \text{ V}) = 101.75 \text{ mA}$. This discharge current is satisfied by the battery as the standard discharge current at 0.2 C is 380 mA.

5 RF & PCB Layout

5.1 RF Antenna Routing

The RF routing for both the LoRa and GPS antennas has been implemented according to standard high-frequency design practice on a 4-layer FR-4 PCB. The LoRa feed is routed in a straight line to the SMA-J-P-H-ST-EM1 connector, while the GPS feed is kept short and curved toward the A-1JB antenna. Both paths are designed as $50\ \Omega$ controlled-impedance transmission lines in order to minimize return loss and maintain proper signal integrity at the operating frequencies of 868/915 MHz for LoRa and 1575.42 MHz for GPS. The resulting GPS feed geometry is illustrated in Figure 5.1a, while the overall LoRa antenna feed is shown in Figure 5.1b.

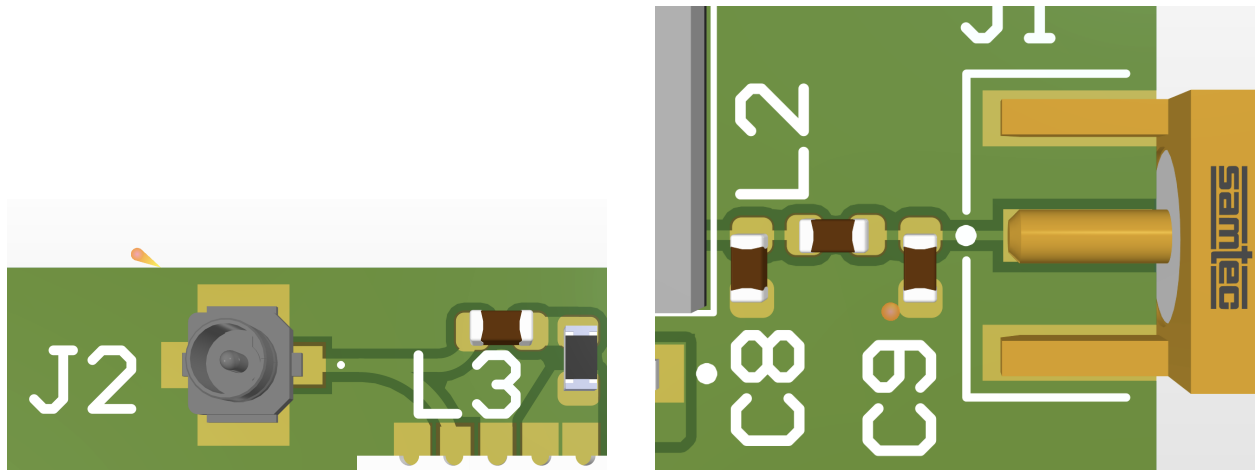
The use of a coplanar waveguide with ground shielding and via stitching is technically appropriate for these RF paths. This structure helps confine the electromagnetic field, improves impedance stability, and reduces susceptibility to external noise and coupling from nearby traces. The ground plane beneath the RF trace must remain continuous and free of splits or interruptions, since any discontinuity in the reference plane can distort the return current path and degrade the effective impedance.

For the RF ground pads, direct connection to the ground pour is preferred over thermal relief connections. Thermal spokes introduce unnecessary inductance into the ground return path, which is undesirable at RF frequencies, particularly at the GPS frequency where even small parasitics can impact matching. Direct connections provide a lower-impedance grounding structure and are therefore better suited for antenna feed regions and other sensitive high-frequency circuitry.

Layer transitions through vias are avoided wherever possible on both the LoRa and GPS feeds. Vias introduce parasitic inductance and create small impedance discontinuities that can reduce matching quality and increase reflections. For this reason, the RF traces in Figures 5.1a and 5.1b are kept on the top layer for the entire antenna feed path.

The curved GPS routing is also preferred over sharp 90-degree corners. While 45-degree mitred bends are acceptable in RF design, curved arcs provide a smoother geometry and reduce local impedance perturbations even further. At 1575.42 MHz, the electrical effect of a bend is usually small if the trace is short and well-controlled, but rounded routing as implemented in Figure 5.1a remains the cleaner solution from a signal-integrity perspective.

Overall, the selected routing strategy follows established RF design principles by preserving a controlled $50\ \Omega$ impedance, limiting discontinuities, reducing EMI, and maintaining a compact layout. This is particularly important for a battery-powered tracking device, where reliable antenna performance and low-loss RF routing directly affect system range and overall effectiveness.



(a) GPS antenna feed routing showing the short, curved $50\ \Omega$ transmission line from the module output to the A-1JB antenna pad.

(b) LoRa antenna feed routing with a straight $50\ \Omega$ controlled-impedance path to the SMA-J-P-H-ST-EM1 connector.

Figure 5.1: RF antenna feed implementations for GPS (left) and LoRa (right).

Power Distribution The 3.3 V rail is routed as a major power backbone directly from the power-management IC, rather than daisy-chained from load to load. This star-like distribution topology reduces resistive voltage drops and minimizes the risk of local supply dips when multiple circuits draw transient current simultaneously. Sensitive RF and analog blocks are therefore fed from a low-impedance common node instead of being placed at the end of a long queue of series segments, which improves overall supply integrity as illustrated in Figure 5.2.

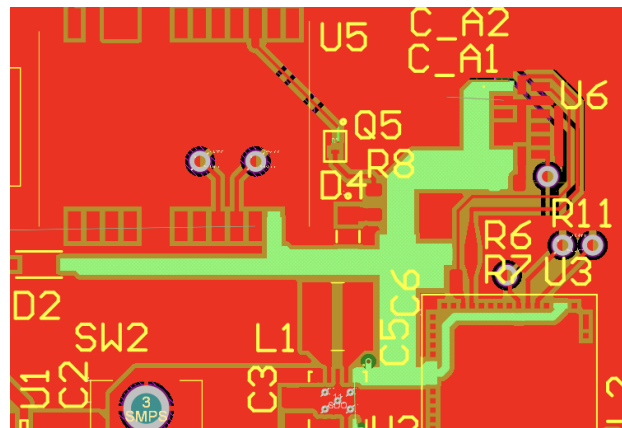


Figure 5.2: Conceptual 3.3 V rail distribution where all major loads are fed directly from the power IC rather than in a daisy-chain.

6 Power Budget & Battery Life

This chapter looks at the power budget of the bike tracker. We will compile the current draw of all relevant components in the design, into a table to find the current draw in the span of a day, and subsequently, the battery life. To do this we will define three power states for the bike tracker.

- **Sleep mode** The bike has been parked and the device is armed, but has not detected any movement. All components will be in sleep mode.
- **GPS fix** The device has detected a theft, is waking the MCU and attempting to get a GPS fix. A hot start is assumed.
- **LoRa TX** The device now has a GPS fix and will transmit coordinates over LoRa.

In the sleep mode the MCU will be in standby mode, which is the lowest sleep mode available while still being able to be woken by hardware interrupts. The RTC and register retention of the MCU will be disabled as they are not needed.

For the GPS we will use ICC tracking metric in power save mode. The backup battery discharges when the device is off and charges when it is on. We assume the battery is charging in the sleep mode and is fully charged in the other modes.

To save power, the accelerometer will operate at low output data rate, as none of the data will be read, only the interrupt is used. The SMPS outputs about 12 mA in this mode, which corresponds to an efficiency of around 75% at 3.6 V. See figure 6.1.

$$I_{SLEEP} = \frac{3.3 \text{ V} \cdot 12 \text{ mA}}{0.75 \cdot 3.6 \text{ V}} = 14.7 \text{ mA} \quad (6.1)$$

Waking the MCU from standby mode with a wake up clock of 8 MHz uses 19.6 nAs (5.44 pAh), which is very low, so it will be excluded from the calculations. The MCU will run at 8 MHz in run mode.

The GPS will perform a hot start as it has the backup battery. This is estimated to take about 15 seconds in the datasheet, but we will assume 1 minute. The SMPS outputs 48 mA in GPS Fix mode, which is about 85% in efficiency.

We assume the bike is recovered 10 hours after being reported stolen by the device. This time span is likely to vary a lot in practice. All currents can be seen in table 6.1.

$$I_{GPS \text{ FIX}} = \frac{3.3 \text{ V} \cdot 48 \text{ mA}}{0.75 \cdot 3.6 \text{ V}} = 51.8 \text{ mA} \quad (6.2)$$

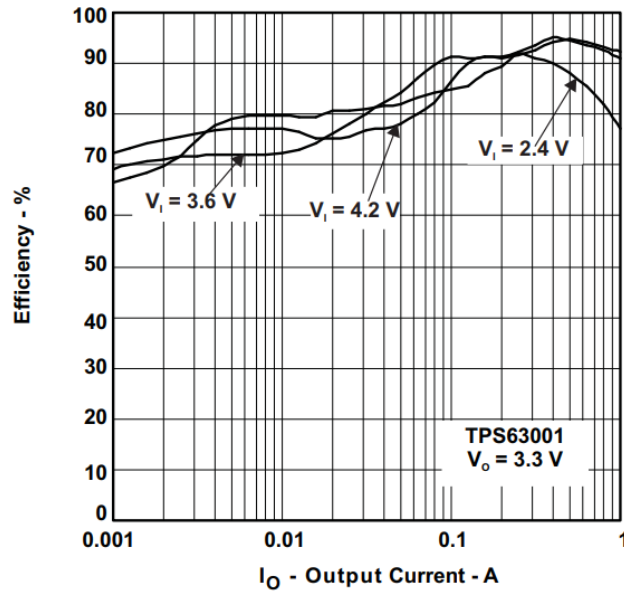


Figure 6.1: Efficiency of the SMPS versus the output current.

	Sleep mode	GPS Fix	LoRa TX
MCU	125 nA	1.1 mA	1.1 mA
GPS	11 mA	47 mA	11 mA
Backup Battery	879 μ A	0 mA	0 mA
Accelerometer	30 μ A	30 μ A	30 μ A
SMPS output	12 mA	48 mA	12 mA
Battery discharge	14.7 mA	51.8 mA	14.7 mA
Time spent in mode	14 h/day	1 m/day	10 h/day

Table 6.1: Current draw of each component in each mode.

The battery supplies 14.7 mA pretty much all day except for a single hot start of the GPS about every day. Now the calculations for battery drain and battery life can be done.

$$14.7 \text{ mA} \cdot 24 \frac{\text{h}}{\text{day}} + 51.8 \text{ mA} \cdot 1/60 \frac{\text{h}}{\text{day}} = 353.7 \frac{\text{mAh}}{\text{day}} \quad (6.3)$$

$$\frac{1900 \text{ mAh}}{353.7 \text{ mAh/day}} = 5.4 \text{ days} \quad (6.4)$$

The battery will last approximately 5 days, but this is at max use, so in practice the device likely only needs charging once a week.